Program Description: Numeric Processing

In this assignment, we will be utilizing functions to read and write data to files. Attached to the assignment is the file “Random.txt”. This file contains an unspecified amount of random numbers, with each number entered on its own line.

Behavior:

Your task is to create separate functions to perform the following operations:

1. Read all numbers from the file and display them to the screen
2. Count the amount of numbers stored in the file and display to the screen
3. Calculate the sum of all numbers in the file and display to the screen
4. Calculate the average of all numbers in the file and display to the screen
5. Write the results from the prior functions (amount, sum, average) to a separate file named “Result.txt”

Each function should be executed in the order listed above at runtime, without the need for user interaction.

You must also utilize exception handling techniques to handle at least one appropriate exception in this program.

The structure and overall design of each function is left up to you, as long as the intended functionality is accomplished.

Also be sure to implement docstrings for each function to provide proper documentation.

Example Output:

```
Reading from file Random.txt
1
2
3
4
5
There are 5 numbers within the file
The sum of all numbers in the file is 15
The average of all numbers in the file is 3.
```

```
The results have been written to file Results.txt
```

Constraints:

a) Make sure you use the following comment at the beginning of your program:

```
# Homework 5:      Numeric Processing
# Program By:      YOUR NAME
# File Name:       H5_Num_Process.py
# Function:        This program reads and writes to files
```

b) **ATTACH** your completed .py file to the *Assignment Dropbox*. Do not cut and paste your code into the message section.